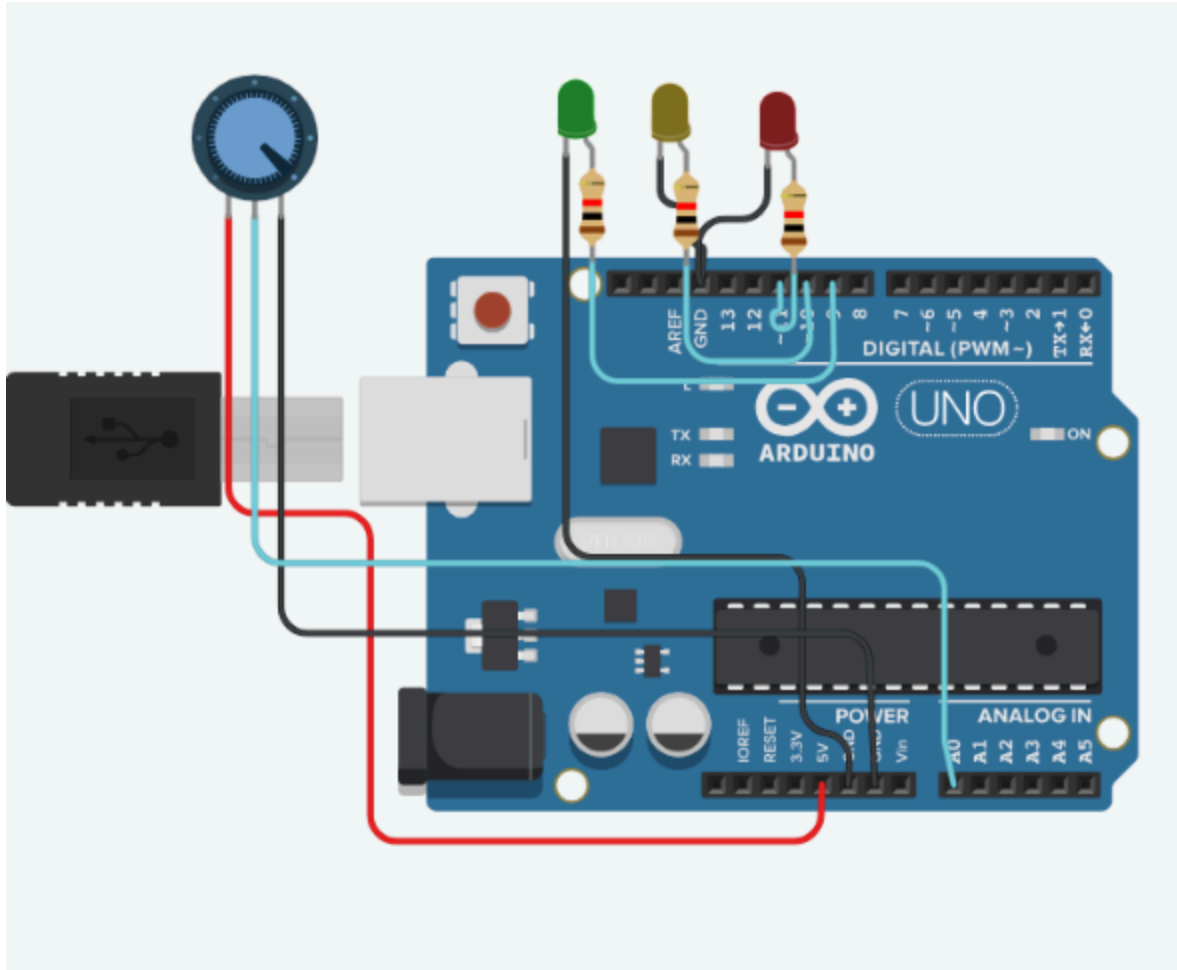




```
int green = 9;
int yellow = 10;
int red = 11;

int pw;
int brightness;

void setup()
{
  pinMode(green, OUTPUT);
  pinMode(yellow, OUTPUT);
  pinMode(red, OUTPUT);

  Serial.begin(9600);
}

void loop()
```

```

{
  pw = analogRead(A0);

  if(pw >= 0 && pw <= 340)
  {
    brightness = map(pw, 0, 340, 0, 255);

    analogWrite(green, brightness);
    digitalWrite(yellow, LOW);
    digitalWrite(red, LOW);

    Serial.print("Potentiometer: ");
    Serial.print(pw);
    Serial.print(" | LED State: Green ON, Yellow OFF, Red OFF");
    Serial.print(" | Brightness: ");
    Serial.println(brightness);
  }

  else if(pw >= 341 && pw <= 680)
  {
    digitalWrite(green, LOW);
    analogWrite(yellow, 255);
    digitalWrite(red, LOW);

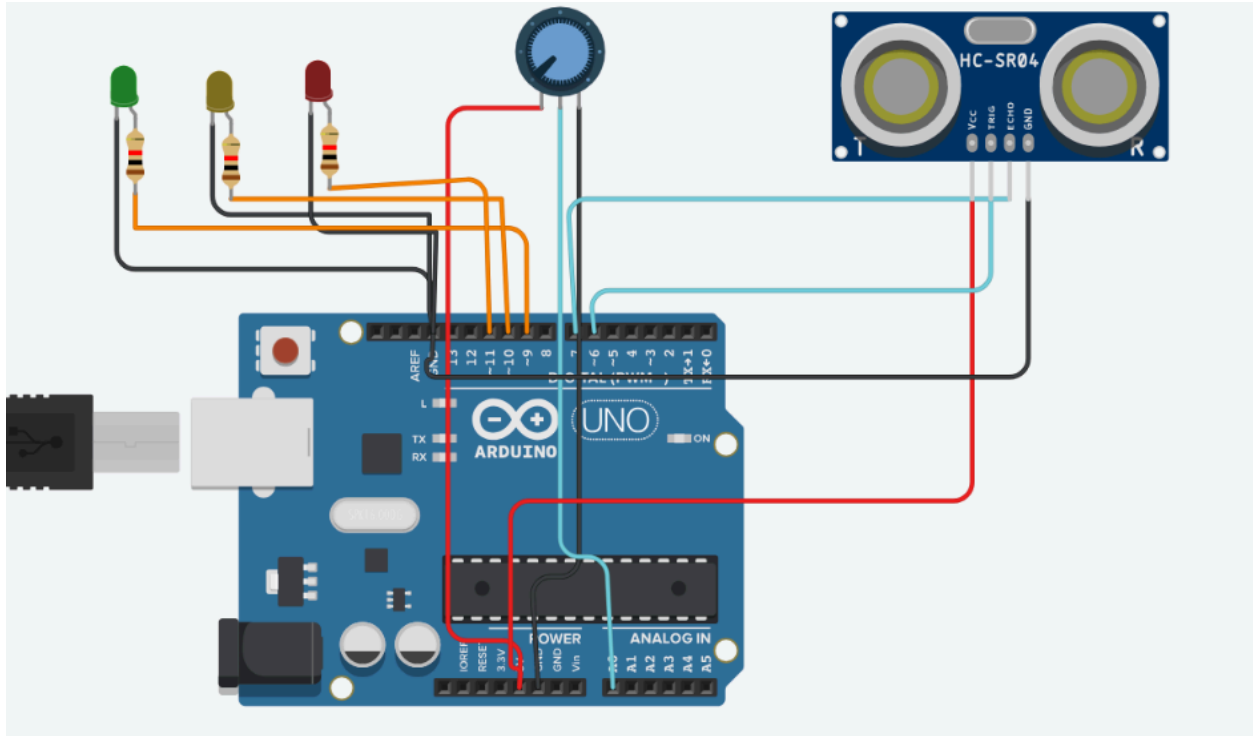
    Serial.print("Potentiometer: ");
    Serial.print(pw);
    Serial.println(" | LED State: Green OFF, Yellow ON, Red OFF");
  }

  else if(pw >= 681 && pw <= 1023)
  {
    digitalWrite(green, LOW);
    digitalWrite(yellow, LOW);
    analogWrite(red, 255);

    Serial.print("Potentiometer: ");
    Serial.print(pw);
    Serial.println(" | LED State: Green OFF, Yellow OFF, Red ON");
  }
}

```

Q.2



```
int green = 9;  
int yellow = 10;  
int red = 11;
```

```
int pw;  
int trig = 6;  
int echo = 7;
```

```
int us;  
long duration;  
float warn;
```

```
float myMap(float value, float inMin, float inMax, float outMin, float outMax)  
{  
    return (value - inMin) * (outMax - outMin) / (inMax - inMin) + outMin;  
}
```

```
void setup()  
{  
    pinMode(green, OUTPUT);  
    pinMode(yellow, OUTPUT);  
    pinMode(red, OUTPUT);
```

```

pinMode(trig, OUTPUT);
pinMode(echo, INPUT);

Serial.begin(9600);
}

void loop()
{
  digitalWrite(trig, LOW);
  delayMicroseconds(2);

  digitalWrite(trig, HIGH);
  delayMicroseconds(10);
  digitalWrite(trig, LOW);

  duration = pulseIn(echo, HIGH);

  us = duration * 0.034 / 2;

  pw = analogRead(A0);

  warn = myMap(pw, 0, 1023, 10, 50);

  Serial.print("Potentiometer: ");
  Serial.print(pw);

  Serial.print(" | Warning Distance: ");
  Serial.print(warn);

  Serial.print(" cm | Current Distance: ");
  Serial.print(us);

  if(us <= warn)
  {
    digitalWrite(green, LOW);
    digitalWrite(yellow, LOW);
    digitalWrite(red, HIGH);

    Serial.println(" cm | Status: RED - WARNING");
  }

  else if(us <= 2 * warn)
  {
    digitalWrite(green, LOW);

```

```
digitalWrite(yellow, HIGH);  
digitalWrite(red, LOW);  
  
Serial.println(" cm | Status: YELLOW - CAUTION");  
}  
  
else  
{  
    digitalWrite(green, HIGH);  
    digitalWrite(yellow, LOW);  
    digitalWrite(red, LOW);  
  
    Serial.println(" cm | Status: GREEN - SAFE");  
}  
  
delay(500);  
}
```